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PATHOVISION: A CONVOLUTIONAL NEURAL NETWORK APPROACH FOR HISTOPATHOLOGY IMAGE – BASED COLORECTAL CANCER DETECTION WITH WEB APPLICATION INTEGRATION FOR CLINICAL DECISION SUPPORT

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Colorectal cancer continues to be one of the most common malignancies diagnosed globally and represents a significant global health burden [1]. Because cellular morphology and tissue architecture provide information needed to identify malignant alterations, histopathological analysis of tissue is a crucial part of cancer diagnosis [2][3]. However, a variety of cellular and architectural characteristics must be examined in order to explain the complex visual patterns shown in histological images. This has spurred more study into computational techniques that can help in pathological picture analysis.

In this regard, artificial intelligence—especially deep learning—has shown promise [4][5]. **Convolutional Neural Networks** (CNNs) are useful for spotting patterns that could differentiate various tissue types since they can immediately learn hierarchical visual representations from photos [4][5]. Tumor classification, gland segmentation, tumor-microenvironment characterization, and prognosis prediction were identified as the main research areas in a systematic review of AI applications in colorectal pathology. However, limitations in dataset quality and scale for clinical application were also noted [2].

As a result, **deep learning** has grown in importance in colorectal histology. These techniques have been studied for diagnosis, classification [6], molecular characteristic prediction [7], and analysis of clinically important pathological aspects [8], according to a comprehensive review that focused on deep learning [9]. CNN architectures, such as **ResNet, VGG, DenseNet, and EfficientNet**, are still being developed for colorectal cancer image classification, according to more recent reviews [10]-[13]

Despite these advances, an important research gap remains. A model's ability to perform well on a certain dataset does not guaranty that it will translate to photographs taken under various circumstances [14],[15]. **Model performance** can be impacted by discrepancies between training and real-world photos, staining variation, image capture, class imbalance, and dataset size [14],[19]. High overall sensitivity and discrimination for deep-learning colorectal cancer detection were found in a 2026 systematic review and diagnostic meta-analysis [19].

However, it was observed that the evidence does not yet constitute a mature, independently replicated evidence base and is still dominated by a small number of validation situations. Moreover, a forecast by itself offers little insight into the reasoning behind a model's result [20]-[22]. By visualizing the regions that contribute to a CNN's prediction, explainable-AI approaches like **Grad-CAM** may make it possible to analyze model behavior in addition to its output [20],[21]. Explainability, dataset scarcity, standardization, and clinical integration are identified as ongoing research problems in recent reviews [3],[23].

Therefore, this study will look at whether a CNN can classify colorectal histological images as well as whether its practical deployment, interpretability, and performance can all be assessed simultaneously. In order to investigate the possibilities and constraints of incorporating computational image analysis into a practical workflow, the final system will be positioned as clinical decision support rather than an independent diagnostic substitute[19], [23], [28].

Research Questions

How effectively can a Convolutional Neural Network: **PATHOVISION** classify colorectal cancer from histopathological image datasets, and how can its predictions and outcomes be integrated into an interpretable web-based system for assist during clinical procedures?

Three supporting questions will be addressed by the investigation:

a. When tested on previously unknown and untrained photos, how well can **PATHOVISION** classify the chosen colorectal histology categories?

b. Can Grad-CAM offer helpful visual explanations of **PATHOVISION's** predictions, and which areas of the images specifically contribute to these predictions?

c. How well can a web application for picture submission, categorisation, and decision-support visualisation include the trained model and its explanatory outputs?

These questions connect the computational experiment with the practical objective of developing an interpretable system rather than evaluating classification accuracy in isolation.

Literature Review

Deep learning has replaced traditional image-analysis methods in computational colorectal pathology research. Early reviews found that AI applications were mostly used in tumour classification and gland segmentation, but they also pointed out that routine clinical deployment was hampered by the small size and poor quality of many datasets [2][5][3].

35 The ability of CNNs' convolutional layers to automatically extract spatial and morphological data from images led to their later rise to prominence [4], [5]. In their evaluation of deep learning in colorectal histopathology, Davri et al. found uses that go beyond straightforward diagnosis, such as molecular prediction and pathological feature characterisation [6]. ResNet and other transfer-learning designs are often examined in more recent studies that compare CNN architectures for colorectal cancer classification [7], [24].

However, the context of **dataset construction and validation** must be taken into consideration when interpreting reported performance. Promising sensitivity and specificity were observed in the 2026 diagnostic meta-analysis [19], however it was noted that specificity varied 36 depending on endpoint definitions and analytic units, and that a small number of development programs provided a large portion of the data [25]. This illustrates the difference between robust generalisation across clinical situations and excellent performance inside a controlled sample.

Explainability offers an additional area for improvement. While recent reviews highlight explainable AI as a crucial element of computer-assisted colorectal cancer analysis [21], Grad-CAM and related techniques can detect picture regions linked with model predictions [20], [22]

Consequently, the literature demonstrates that CNN-based colorectal histopathology classification is feasible while providing a chance to explore the interplay between classification **performance, generalisation, interpretability, and practical deployment** [6]

– [9]. By combining these components into a single experimental system, this study fills that gap [9], [23].

Methodology

a. Research Design

An **experimental machine-learning** design comprising five stages—dataset preparation, CNN development, model evaluation, explainability analysis, and web application integration—will be used in the study [5], [23], [28].

37 Labelled colorectal histopathology photos will be utilised to train the model, and data not used during training will be used for evaluation. The trained model will be linked to an interface in the final application so that a user can submit an image and see the classification and explanation that results [14], [19], [35].

Dataset and Preprocessing

8 picture classes from a tagged colorectal histopathology dataset will be used in the investigation. To ensure a uniform format for model processing, images will be represented as **150 x 150 RGB inputs**. Images will be sorted by class and examined for corrupted or inappropriate samples prior to training [6].

28 The dataset will be separated into subsets **for testing, validation, and training**. Model parameters will be optimised using the training set, model development will be supported by the validation set, and the testing set won't be viewed until the final assessment. The goal of this division is to lessen data loss and offer a more accurate performance assessment. [14], [19], [35]

Before being sent to CNN, images will be resized and normalised. To add controlled variation and lessen overfitting, **data augmentation** will be performed to training images. Rotation, flipping, scaling, and other adjustments that don't change the underlying diagnostic category are examples of appropriate transformations [15].

b. CNN Development

1. An initial visual feature representation will be provided by a pretrained architecture, and its classification component will then be tailored to the eight target classes. A suitable optimiser and categorical classification loss will be used to train the model.
2. The main experimental implementation will look at ResNet-based classification; 39 32 the validation dataset will be used to track performance during training. To guarantee reproducibility, training parameters including learning rate, batch size, epochs, and regularisation will be recorded.
3. Alternative CNN designs may be assessed under comparable experimental settings when computationally possible. Instead of presuming that one architecture is ideal in advance, this will enable the research to ascertain whether variations in

architecture affect categorisation performance.

c. **Model Evaluation**

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1. While precision, recall, specificity, and F1-score offer class-sensitive performance metrics, accuracy will gauge total correct categorisation. To determine which classes are most commonly confused, a confusion matrix will be employed [32].
2. ROC curves and AUC will offer an extra classification discrimination metric when applicable [34].
3. Images that have been incorrectly classified will also be analysed to find potential reasons for the error, such as morphological similarities, inadequate class representation, poor image quality, or visually unclear structures [6], [14], [16].

d. **Explainability Using Grad – CAM**

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Selected test images will be subjected to Gradient-weighted Class Activation Mapping (Grad-CAM) [20]. Grad-CAM creates a heatmap that highlights areas that significantly influence a model's prediction. To ascertain whether the model's focus appears to be focused on significant tissue regions or possibly unimportant picture aspects, instances that were properly and mistakenly classified will be compared. Beyond numerical performance, this offers an extra level of examination. But rather than being proof of clinical reasoning, the heatmaps will be regarded as interpretive evidence.

e. **Web Application Integration: PATHOVISION**

A web-based tool intended for clinical decision support will thereafter incorporate the trained CNN. With the suggested workflow, a user will be able to:

1. Upload a histopathological image;
2. Validate and preprocess the input;
3. Pass the image to the trained CNN;
4. Display the predicted class and confidence;
5. Generate the corresponding Grad-CAM visualization; and
6. Present the prediction and visualization within a single interface.

PATHOVISION will clearly discern between a medical diagnosis and an algorithmic forecast. Its primary objective is to show how a computer model could offer an extra layer of data to aid in human interpretation.

f. System Evaluation

Valid and invalid image inputs will be tested, along with a. image processing, b. model inference, c. prediction display, d. confidence reporting, and e. Grad-CAM generation. Following that, the entire process—from uploading images to producing the final product—will be tested. When possible, response time and application problems will also be documented [35]

Research Significance and Expected Outcomes

Because it treats colorectal cancer detection as both a machine-learning and a translation challenge, the proposed study is noteworthy. While there is a lot of promise for CNN-based histopathology analysis in the literature now under publication, there are also issues with dataset quality, generalisation, explainability, and clinical application [2], [19], [23], [28].

PATHOVISION, a trained CNN with the ability to categorise the chosen colorectal histopathology categories and a quantitative evaluation of its performance on unseen images, is the anticipated result. Additionally, the study will create an integrated web application that can provide forecasts and visual explanations, as well as Grad-CAM visualisations to examine model behaviour. Crucially, the study will not make any assumptions about its correctness. Rather, the model's efficacy and limitations will be determined by experimental outcomes. This allows errors and weaknesses to form part of the research findings rather than being treated simply as failures of the system. [19], [35].

Limitations and Conclusion

The quantity and representativeness of the dataset used in this study may have limitations. It's possible that performance measures and explanatory findings from a certain dataset won't apply to pictures from other labs, scanners, staining techniques, or patient groups. [14], [16], [17]

Similar to this, recent research points to dataset heterogeneity and a lack of external validation as significant obstacles to clinical application

The number of architectural and hyperparameter combinations that can be studied may also be restricted by computational resources. Furthermore, Grad-CAM cannot independently show clinical validity and only approximates model attention. [20], [22].

11 The overall goal of this study is to determine whether PATHOVISION can reliably classify colorectal histology while also looking at the extra needs of practical deployment and interpretability. The project intends to show how AI-based image analysis could serve as a supportive computational layer in pathology by integrating model evaluation with Grad-CAM and a web-based interface, while acknowledging that clinical diagnosis necessitates suitable expert monitoring [19], [23], [28].

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